

STAR FORMATION

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STARS ARE FORMED by the gravitational collapse of cool, dense interstellar clouds. These clouds are composed mainly of molecular hydrogen (see p.228). A cloud has to be of a certain mass for gravitational collapse to occur, and a trigger is needed for the collapse to start, as the clouds are held up by their own internal pressure. Larger clouds fragment as they collapse, forming sibling protostars that initially lie close together—some so close that they are gravitationally bound. The material heats up as it collapses until, in some clouds, the temperature and pressure at their centers become so great that nuclear fusion begins and a star is born.



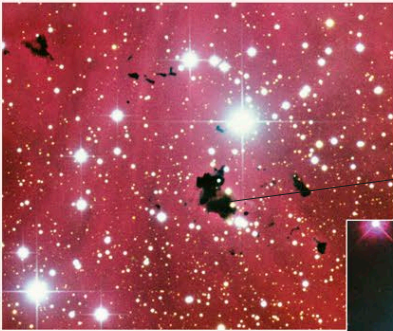
STAR-FORMING REGION
In the nebula RCW 120, in the southern Milky Way, an expanding bubble of ionized gas is causing the surrounding material to collapse into dense clumps, in which new stars will be born.

STELLAR NURSERIES

As well as being among the most beautiful objects in the universe, star-forming nebulae contain a combination of raw materials that makes star birth possible. These clouds of hydrogen molecules, helium, and dust can be massive systems hundreds of light-years across or smaller individual clouds, known as Bok globules. Although they may lie undisturbed for millions of years, disturbances can trigger these nebulae to collapse and fragment into smaller clouds from which stars are formed. Remnants from the star-forming nebulae will surround the stars, and the stellar winds produced by the new stars can, in turn, cause these remnants to collapse. If the clouds are part of a larger complex, this can become a great stellar nursery. Massive stars have relatively short lives, and they can be born, live, and die as a supernova while their less-massive siblings are still forming. The shock wave from the supernova may plow through nearby interstellar matter, triggering yet more star birth.

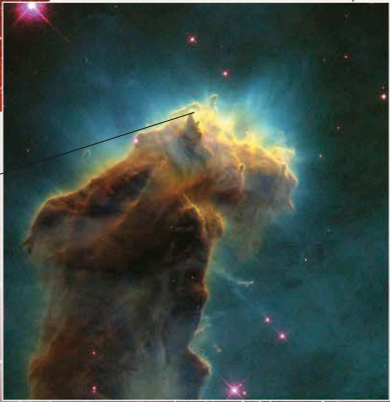
FORMATION IN ACTION

This spectacular vista captures the outstretched wings of the so-called Seagull Nebula, a star-forming region in the constellation of Monoceros. Young, bright stars are still surrounded by glowing nebulosity, while dark clouds of denser gas and dust silhouetted against the glowing background mark locations where star birth is still continuing.



BOK GLOBULE
Small, cool clouds of dust and gas, known as Bok globules, are the origins of some of the Milky Way's lower-mass stars.

Bok globule



stellar EGGS

STELLAR EGGS

Within the evaporating gaseous globules (EGGS) of the Eagle Nebula, interstellar material is collapsing to form stars.